

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $\bar{w} > \bar{w}_s$, $b > 0$, $p > 0$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}(\bar{w} - \bar{w}_s, b, p) = 1 - \left(1 + \frac{\bar{w} - \bar{w}_s}{b}\right)^{-p} \quad (\bar{w}_s - 1 : \text{unknown surface position})$$

$$\bar{a}(\bar{w} - \bar{w}_s, b_w, p_w) = 1 - \left(1 + \frac{\bar{w} - \bar{w}_s}{b_w}\right)^{-p_w}$$

$$\bar{a}(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = 1 - \left(1 + \frac{\bar{w}_d - \hat{w}_s}{\hat{b}_w}\right)^{-\hat{p}_w}$$

$$a_w = a_o \bar{a}(\bar{w} - \bar{w}_s, b_w, p_w), \quad a_o = \frac{\Delta t}{\gamma_N R} \quad (a_o \text{ fixed, not a state})$$

$$p = 1 : \quad \bar{a} = \frac{\bar{w} - \bar{w}_s}{(\bar{w} - \bar{w}_s) + b}, \quad \frac{d\bar{a}}{d\bar{w}} = \frac{b}{((\bar{w} - \bar{w}_s) + b)^2}$$

2 Gain rate and parameter Jacobians

$$\bar{a}'(\bar{w} - \bar{w}_s, b, p) := \frac{d\bar{a}}{d\bar{w}} = \frac{p}{b} \left(1 + \frac{\bar{w} - \bar{w}_s}{b}\right)^{-p-1}$$

$$\bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = \frac{\hat{p}_w}{\hat{b}_w} \left(1 + \frac{\bar{w}_d - \hat{w}_s}{\hat{b}_w}\right)^{-\hat{p}_w-1}$$

$$\frac{\partial \bar{a}}{\partial \bar{w}_s} = -\bar{a}' \quad (\text{exact, any } b, p)$$

$$\begin{aligned} & \bar{a}'(\bar{w}_d - \bar{w}_s, b_w, p_w) - \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) \\ &= J_b(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) e_{b_w} + J_p(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) e_{p_w} + J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) e_{\bar{w}_s} \end{aligned}$$

$$J_b(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = \left[-\frac{1}{\hat{b}_w} + \frac{(\hat{p}_w + 1)(\bar{w}_d - \hat{w}_s)}{\hat{b}_w((\bar{w}_d - \hat{w}_s) + \hat{b}_w)} \right] \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w)$$

$$J_p(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = \left[\frac{1}{\hat{p}_w} - \ln\left(1 + \frac{\bar{w}_d - \hat{w}_s}{\hat{b}_w}\right) \right] \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w)$$

$$J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = \left[\frac{\hat{p}_w + 1}{(\bar{w}_d - \hat{w}_s) + \hat{b}_w} \right] \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w)$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d+1, \quad d = 2; \quad \delta \bar{w}_D \equiv 0 \text{ hereafter})$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{a}_w[k+1] &= \bar{a}_w(\bar{w}[k+1]) = \bar{a}(\bar{w}_d[k+1] - \bar{w}_s - \delta \bar{w}_3[k+1], b_w, p_w) \\ \bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \end{aligned}$$

$$\begin{aligned} \delta \bar{w}_1[k+1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k+1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \quad (e_{\bar{a}_w} = \bar{a}_w - \hat{\bar{a}}_w) \\ b_w[k+1] &= b_w[k] \\ p_w[k+1] &= p_w[k] \\ \bar{w}_s[k+1] &= \bar{w}_s[k] \end{aligned}$$

$$\begin{aligned} F_{dw}[k] &:= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\ \varepsilon_w[k] &:= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] + (1 - \lambda_c) n_w[k] \end{aligned}$$

4 Process model

$$\bar{w}[k+1] - \bar{w}[k] = \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]$$

$$\begin{aligned} \bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ \bar{a}'[k] &= \bar{a}'(\bar{w}_d[k] - \bar{w}_s[k], b_w[k], p_w[k]) \end{aligned}$$

$$\mathbf{x}[k] = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b_w, p_w, \bar{w}_s]^\top[k] \quad (d = 2)$$

$$\begin{aligned} \delta \bar{w}_1[k+1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k+1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ b_w[k+1] &= b_w[k] \\ p_w[k+1] &= p_w[k] \\ \bar{w}_s[k+1] &= \bar{w}_s[k] \end{aligned}$$

5 Estimator

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + \hat{a}'[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] \\
\hat{b}_w[k+1] &= \hat{b}_w[k] \\
\hat{p}_w[k+1] &= \hat{p}_w[k] \\
\hat{w}_s[k+1] &= \hat{w}_s[k] \\
\hat{a}'[k] &= \bar{a}'(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k])
\end{aligned}$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$e_{\delta \bar{w}_j} = \delta \bar{w}_j - \delta \hat{w}_j \quad (j = 1, 2, 3), \quad \mathbf{e} = [e_{\delta \bar{w}_1}, e_{\delta \bar{w}_2}, e_{\delta \bar{w}_3}, e_{\bar{a}_w}, e_{b_w}, e_{p_w}, e_{\bar{w}_s}]^\top$$

$$\begin{aligned}
e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\
e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\
e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= [1 + \hat{a}'[k] F_{dw}[k]] e_{\bar{a}_w}[k] + J_{b_w}[k] [\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k]] e_{b_w}[k] \\
&\quad + J_{p_w}[k] [\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k]] e_{p_w}[k] + J_{w_s}[k] [\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k]] e_{\bar{w}_s}[k] \\
&\quad + (1 - \lambda_c) \hat{a}'[k] e_{\delta \bar{w}_3}[k] + \hat{a}'[k] \varepsilon_w[k] \\
e_{b_w}[k+1] &= e_{b_w}[k] \\
e_{p_w}[k+1] &= e_{p_w}[k] \\
e_{\bar{w}_s}[k+1] &= e_{\bar{w}_s}[k]
\end{aligned}$$

$$\begin{aligned}
J_{b_w}[k] &= J_b(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k]) \\
J_{p_w}[k] &= J_p(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k]) \\
J_{w_s}[k] &= J_{w_s}(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k])
\end{aligned}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 & 0 & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}' & 1+\hat{a}'F_{dw} & J_{b_w}[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{w}_3] & J_{p_w}[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{w}_3] & J_{w_s}[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{w}_3] \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = [0, 0, -\varepsilon_w[k], \hat{a}'[k]\varepsilon_w[k], 0, 0, 0]^\top \quad (q_4 = -\hat{a}'[k]q_3)$$

7 Measurement and innovation

Measurement (d -step delay; $\bar{a}_{wm} = a_{wm}/a_o$; y_2 raw readout — AR(1) whitening enters with R_2 , S8)

$$\begin{aligned} \nabla_d \bar{w}_d[k] &:= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_m[k] = \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_{wm}[k] = \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

Delay back-off

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'[k] \nabla_d \bar{w}_d[k] \quad \implies \quad y_2[k] = \bar{a}_w[k] - \bar{a}'[k] \nabla_d \bar{w}_d[k] + n_a[k]$$

Linearization (partials at the estimate; $\bar{a}'$ does not depend on the state $\bar{a}_w$, so $\partial y_2 / \partial \bar{a}_w$ is exactly 1)

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1, \quad \frac{\partial y_2}{\partial b_w} = -\nabla_d \bar{w}_d[k] J_{b_w}[k], \quad \frac{\partial y_2}{\partial p_w} = -\nabla_d \bar{w}_d[k] J_{p_w}[k], \quad \frac{\partial y_2}{\partial \bar{w}_s} = -\nabla_d \bar{w}_d[k] J_{w_s}[k]$$

Output equation ($\mathbf{H}$ is the Jacobian; row 2 is nonlinear in $\mathbf{x}$, so $\mathbf{H}\mathbf{x}$ is its linearization, not an identity)

$$\mathbf{y}[k] = \mathbf{H}[k] \mathbf{x}[k] + \mathbf{r}[k], \quad \mathbf{r}[k] = \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\nabla_d \bar{w}_d J_{b_w} & -\nabla_d \bar{w}_d J_{p_w} & -\nabla_d \bar{w}_d J_{w_s} \end{bmatrix}$$

Innovation

$$\begin{aligned}
e_{y_1}[k] &= y_1[k] - \delta \hat{w}_1[k] = e_{\delta \bar{w}_1}[k] + n_w[k] \\
e_{y_2}[k] &= y_2[k] - \hat{a}_w[k] + \hat{a}'[k] \nabla_d \bar{w}_d[k] \\
&= e_{\bar{a}_w}[k] - \nabla_d \bar{w}_d[k] [J_{b_w}[k] e_{b_w}[k] + J_{p_w}[k] e_{p_w}[k] + J_{w_s}[k] e_{\bar{w}_s}[k]] + n_a[k]
\end{aligned}$$

8 Process and measurement noise

Process noise Q (only the current white thermal step enters Q : the past thermal steps in ε_w are cross-step correlated, and n_w already sits in R_1)

$$\begin{aligned}
q_3[k] &= -\bar{a}_w[k] \bar{f}_T[k], & q_4[k] &= \hat{a}'[k] \bar{a}_w[k] \bar{f}_T[k] = -\hat{a}'[k] q_3[k] \\
\gamma(\bar{w}) &= \gamma_N c(\bar{w}), & \sigma_{\bar{f}_T}^2 &= a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}, & \bar{a}_w &= \frac{\Delta t}{a_o \gamma(\bar{w}) R} \implies \text{Var}(\bar{a}_w \bar{f}_T) = \frac{4k_B T a_o}{R} \bar{a}_w
\end{aligned}$$

$$Q_{33}[k] = \frac{4k_B T a_o}{R} \bar{a}_w[k], \quad Q_{34}[k] = Q_{43}[k] = -\hat{a}'[k] Q_{33}[k], \quad Q_{44}[k] = \hat{a}'[k]^2 Q_{33}[k]$$

$$Q_{55} = Q_{66} = Q_{77} = 0 \quad (b_w, p_w, \bar{w}_s \text{ constant})$$

All other entries vanish; the gain block is rank one, since $q_4 = -\hat{a}'[k] q_3$. (run time: entries at the current estimate)

Measurement noise $R = \text{diag}(R_1, R_2)$ (a_{cov} = EWMA coefficient, IF_{var} = MA(d) inflation factor of $\hat{\sigma}_{\delta \bar{w}_r}^2$; y_2 treated as white here — whitened treatment in the ref notes)

$$R_1 = \sigma_{n_w}^2$$

$$\sigma_{\delta \bar{w}_r}^2[k] = C_{dpmr} \frac{4k_B T a_o}{R} \bar{a}_w[k] + C_n \sigma_{n_w}^2 = C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w[k] + \xi)$$

$$\xi := \frac{C_n}{C_{dpmr}} \frac{\sigma_{n_w}^2}{4k_B T a_o / R}$$

$$\bar{a}_{wm}[k] = \frac{\hat{\sigma}_{\delta \bar{w}_r}^2[k] - C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}$$

$$\text{Var}(\hat{\sigma}_{\delta \bar{w}_r}^2) = K_{var} IF_{var} (\sigma_{\delta \bar{w}_r}^2)^2, \quad K_{var} = \frac{2a_{cov}}{2 - a_{cov}}$$

$$R_2[k] = K_{var} IF_{var} (\bar{a}_w[k] + \xi)^2 + d Q_{44}[k]$$

(C_{dpmr} , C_n : `Cdpmr_Cn_derivation.tex` — stationary, noise-only loop assumption: OPEN, defect 2; see ref)